Partial Fractions

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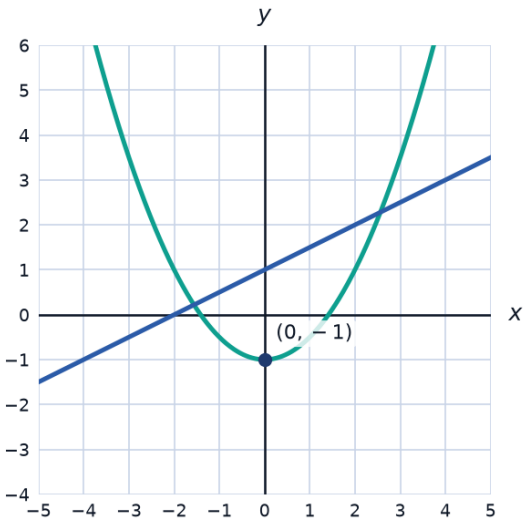
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Distinct linear factors

Express $(5x + 7)/((x + 1)(x + 2))$ in partial fractions.

2 [Foundational] • Target: Distinct linear factors

Express $(3x - 1)/(x(x - 3))$ in partial fractions.



3 [Foundational] • Target: Repeated linear factors

Express $(4x + 1)/(x - 1)^2$ in partial fractions of the form $A/(x - 1) + B/(x - 1)^2$.

4 [Foundational] • Target: Repeated linear factors

Express $(2x^2 - 3x + 1)/(x(x - 1)^2)$ in partial fractions of the form $A/x + B/(x - 1) + C/(x - 1)^2$.

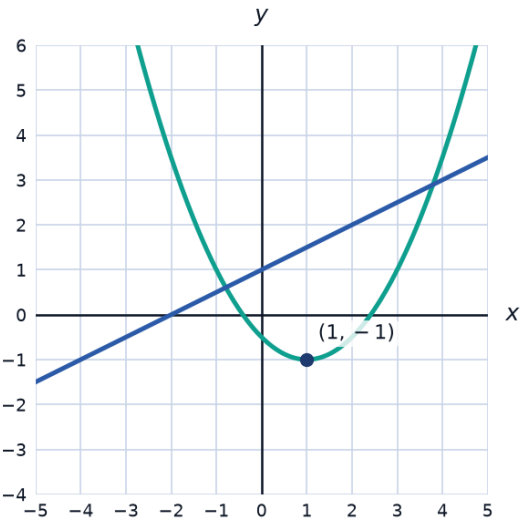
5 [Foundational] • Target: Irreducible quadratic factors

Express $(3x^2 + 2)/((x - 1)(x^2 + 1))$ in partial fractions of the form $A/(x - 1) + (Bx + C)/(x^2 + 1)$.

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6 [Foundational] • Target: Applying partial fractions to integration

Use partial fractions to evaluate $\int (5x + 7)/((x + 1)(x + 2))\,dx$, using the decomposition found earlier in this worksheet.



7 [Foundational] • Target: Applying partial fractions to a binomial series expansion

Express $1/((1 - x)(1 - 2x))$ in partial fractions, then use the result to write the first three terms of its series expansion in ascending powers of x.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Combining terms before decomposing

Express $(7 - x)/((x + 1)(x - 2))$ in partial fractions.

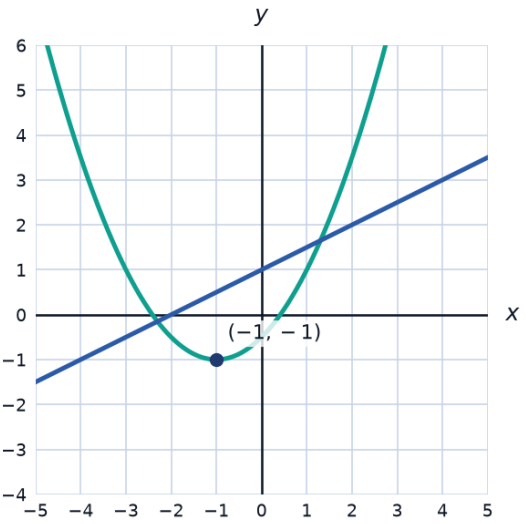
9 [Intermediate] • Target: Combining terms before decomposing

Express $(x^2 + x + 1)/(x(x + 1))$ in partial fractions. (Hint: the degree of the numerator equals the degree of the denominator, so first perform polynomial division.)

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10 [Intermediate] • Target: Applying partial fractions to a rate problem

A population model gives the rate of change $dP/dt = 1/(P(10 - P))$. Express $1/(P(10 - P))$ in partial fractions of the form $A/P + B/(10 - P)$.



11 [Intermediate] • Target: Partial fractions with three distinct linear factors

Express $(6x^2 - 2x - 1)/((x - 1)(x + 1)(x - 2))$ in partial fractions.

12 [Intermediate] • Target: Partial fractions with an improper fraction and repeated factor

Express $x^3/(x - 1)^2$ in partial fractions, first performing polynomial division since the numerator has higher degree than the denominator.

13 [Intermediate] • Target: Using partial fractions to sum a series

Express $1/(n(n + 1))$ in partial fractions, and use the result to find $1/(1 \cdot 2) + 1/(2 \cdot 3) + 1/(3 \cdot 4) + 1/(4 \cdot 5)$ by telescoping.

14 [Intermediate] • Target: Decomposing a fraction with a repeated quadratic-adjacent structure

Express $(x + 5)/((x - 1)(x + 1)^2)$ in partial fractions of the form $A/(x - 1) + B/(x + 1) + C/(x + 1)^2$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

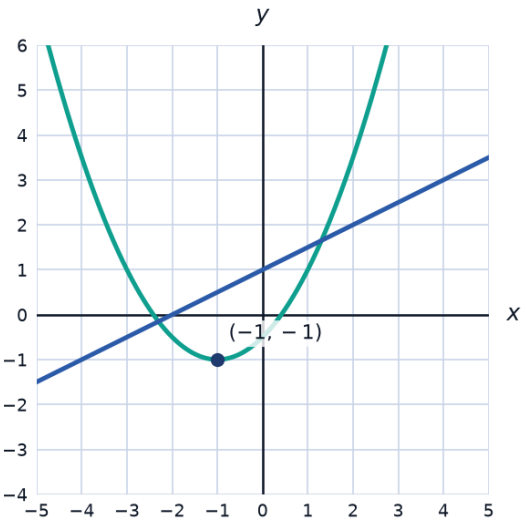
15 [Challenging] • Target: Decomposing a fraction with a repeated quadratic-adjacent structure • Cross-Lesson: 1.01 Proof by Mathematical Induction

Express $(2x^2 + x - 1)/(x^2(x + 1))$ in partial fractions of the form $A/x + B/(x^2) + C/(x + 1)$.

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16 [Challenging] • Target: Decomposing a fraction with a repeated quadratic-adjacent structure • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Use partial fractions to find the Maclaurin (power series) expansion of $1/((1-x)(2-x))$ up to the term in x^2, first decomposing into the form $A/(1-x)+B/(2-x)$.



17 [Challenging] • Target: Decomposing a fraction with a repeated quadratic-adjacent structure • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Express $(4x^2-5x-15)/((x+1)(x-2)(x+3))$ in partial fractions, using the cover-up method (evaluating the numerator at each root while covering the matching factor in the denominator).

18 [Challenging] • Target: Decomposing a fraction with a repeated quadratic-adjacent structure • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

A chemical reaction's rate is modeled by $dC/dt = 1/((C+2)(C-1))$, where C is a concentration. Express $1/((C+2)(C-1))$ in partial fractions of the form $A/(C+2)+B/(C-1)$, and then use the result to write the general antiderivative $\int 1/((C+2)(C-1))\,dC$ in terms of natural logarithms.

19 [Challenging] • Target: Decomposing a fraction with a repeated quadratic-adjacent structure • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Use the decomposition $1/(3x)+8/(3(x-3))$ found earlier in this worksheet for $(3x-1)/(x(x-3))$ to evaluate $\int_4^5 (3x-1)/(x(x-3))\,dx$, giving your answer as a single logarithm expression.

20 [Challenging] • Target: Decomposing a fraction with a repeated quadratic-adjacent structure • Cross-Lesson: 3.01 3D Vectors & the Cross Product

This question has three parts. (a) Factor x^2-4 and hence express $(x+8)/(x^2-4)$ in partial fractions. (b) Use your result from part (a) to write down $\int (x+8)/(x^2-4)\,dx$. (c) Hence evaluate $\int_3^5 (x+8)/(x^2-4)\,dx$, giving your answer as a single expression involving natural logarithms.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Identifying the correct form of a decomposition

Which of the following is the correct form for the partial fraction decomposition of $(3x + 1)/((x - 2)^2(x + 1))$?

a. A/(x-2)+B/(x-2)^2+C/(x+1)

b. A/(x-2)+B/(x+1)

c. A/((x-2)^2)+B/(x+1)

d. (Ax+B)/(x-2)^2+C/(x+1)

22 [Foundational] • Target: Distinct Linear Factors

Decompose $\frac{5x-1}{(x-1)(x+2)}$ into partial fractions $\frac{A}{x-1} + \frac{B}{x+2}$.

a. $A = \frac{4}{3}, B = \frac{11}{3}$

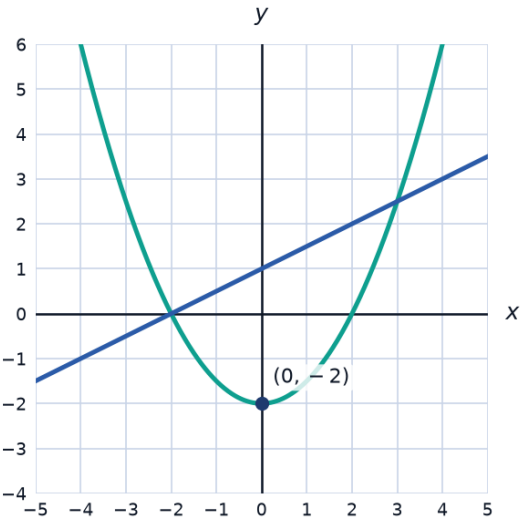
b. $A = 4, B = 11$

c. $A = 1, B = 4$

d. $A = 2, B = 3$

23 [Foundational] • Target: Distinct Linear Factors

Find the partial fraction decomposition of $\frac{1}{x^2 - 9}$.



a. $\frac{1/6}{x-3} - \frac{1/6}{x+3}$

b. $\frac{1/3}{x-3} + \frac{1/3}{x+3}$

c. $\frac{1}{x-3} - \frac{1}{x+3}$

d. $\frac{1/6}{x-3} + \frac{1/6}{x+3}$

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24 [Foundational] • Target: Distinct Linear Factors

Evaluate $\int \frac{3}{(x+1)(x+4)}dx$.

a. $\ln\left|\frac{x+1}{x+4}\right| + C$

b. $3\ln|x+1| + 3\ln|x+4| + C$

c. $\frac{1}{3}\ln\left|\frac{x+1}{x+4}\right| + C$

d. $\ln|(x+1)(x+4)| + C$

25 [Foundational] • Target: Repeated Linear Factors

What is the correct template for decomposing $\frac{2x+7}{(x-2)^2(x+1)}$?

a. $\frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{x+1}$

b. $\frac{A}{x-2} + \frac{B}{x+1}$

c. $\frac{Ax+B}{(x-2)^2} + \frac{C}{x+1}$

d. $\frac{A}{(x-2)^2} + \frac{B}{x+1}$

26 [Foundational] • Target: Repeated Linear Factors

In $\frac{3x^2+1}{(x-1)^2} = A + \frac{B}{x-1} + \frac{C}{(x-1)^2}$, find C.

a. 4

b. 3

c. 1

d. 0

27 [Foundational] • Target: Repeated Linear Factors

Decompose $\frac{x}{(x+1)^2}$.

a. $\frac{1}{x+1} - \frac{1}{(x+1)^2}$

b. $\frac{1}{x+1} + \frac{1}{(x+1)^2}$

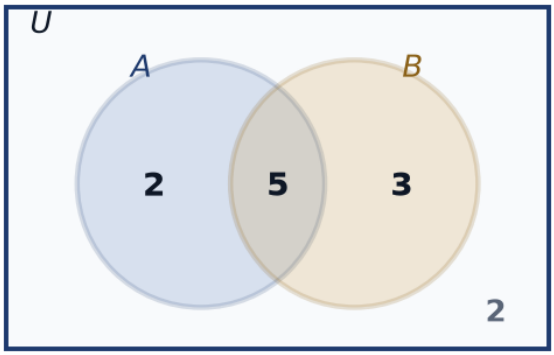
c. $\frac{x}{x+1} - \frac{1}{(x+1)^2}$

d. $\frac{2}{x+1} - \frac{1}{(x+1)^2}$

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28 [Foundational] • Target: Irreducible Quadratic Factors

What partial fraction form should be set up for $\frac{4x^2+2x+5}{(x-3)(x^2+4)}$?



a. $\frac{A}{x-3}+\frac{Bx+C}{x^2+4}$

b. $\frac{A}{x-3}+\frac{B}{x^2+4}$

c. $\frac{A}{x-3}+\frac{B}{x+2}+\frac{C}{x-2}$

d. $\frac{Ax+B}{x-3}+\frac{Cx+D}{x^2+4}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Irreducible Quadratic Factors

In the decomposition $\frac{x^2+1}{(x-2)(x^2+1)}=\frac{A}{x-2}+\frac{Bx+C}{x^2+1}$, find A.

a. $\frac{5}{5}=1$

b. 0

c. 2

d. -1

30 [Intermediate] • Target: Irreducible Quadratic Factors

Evaluate $\int\frac{2x+3}{x^2+9}dx$.

a. $\ln(x^2+9)+\arctan\left(\frac{x}{3}\right)+C$

b. $\ln(x^2+9)+3\arctan(x)+C$

c. $\frac{1}{2}\ln(x^2+9)+\frac{1}{3}\arctan(x/3)+C$

d. $2\ln(x^2+9)+C$

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31 **[Intermediate]** • **Target: Improper Rational Fractions (Long Division First)**

Before performing partial fractions on $\frac{x^3+2x^2+1}{x^2-1}$, what preliminary algebraic step is mandatory?

a. Polynomial long division to separate into polynomial quotient plus proper remainder

b. Factoring the numerator completely

c. Applying the cover-up method directly

d. Multiplying numerator and denominator by x

32 **[Intermediate]** • **Target: Improper Rational Fractions (Long Division First)**

Perform polynomial long division on $\frac{x^3+2x^2+1}{x^2-1}$. What is the quotient?

a. $x+2$

b. $x-2$

c. x^2+2

d. x

33 **[Intermediate]** • **Target: Improper Rational Fractions (Long Division First)**

Decompose the remainder term $\frac{x+3}{x^2-1} = \frac{x+3}{(x-1)(x+1)}$.

a. $\frac{2}{x-1} - \frac{1}{x+1}$

b. $\frac{1}{x-1} + \frac{2}{x+1}$

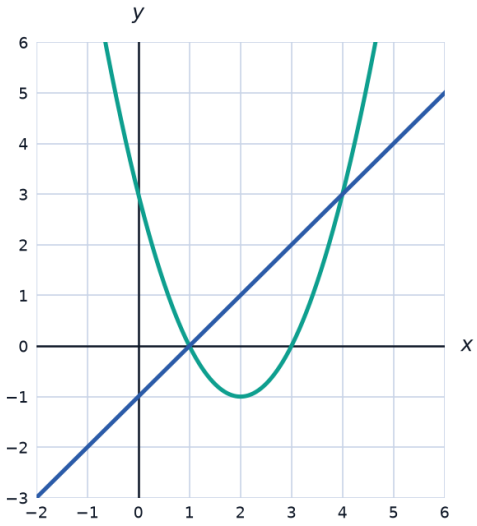
c. $\frac{3}{x-1} - \frac{2}{x+1}$

d. $\frac{1}{x-1} - \frac{1}{x+1}$

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34 **[Intermediate]** • **Target: Solving a separable differential equation using partial fractions**

Consider the quadratic function $f(x) = x^2 - 4x + 3$ and the linear function $g(x) = x - 1$. The points of intersection are given by their solutions. What is the y-coordinate of the point of intersection where x = 1?



a. 0

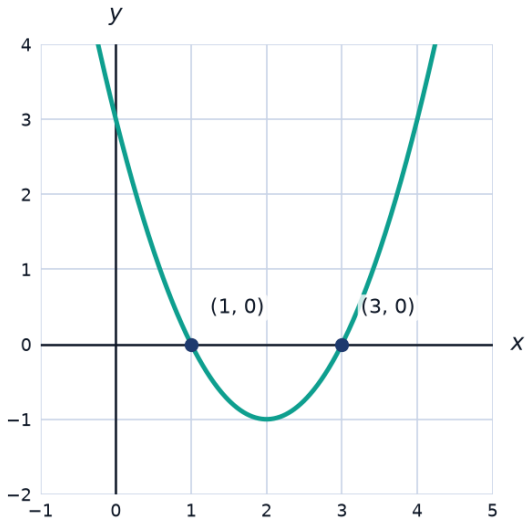
b. 1

c. -1

d. 2

35 **[Intermediate]** • **Target: Explaining when polynomial division is required**

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points where $f(x) = 0$. Determine the sum of the x-coordinates of these intersection points.



a. 1

b. 3

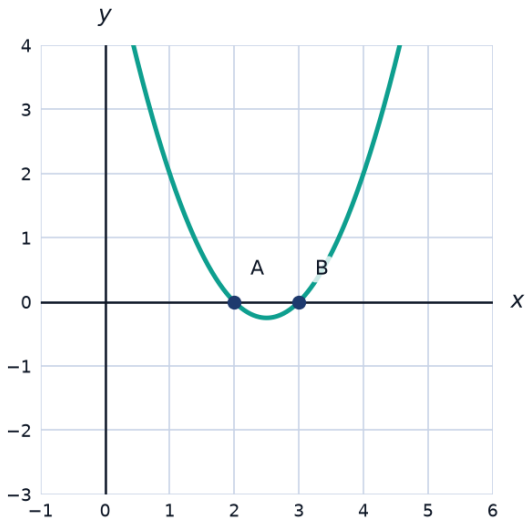
c. 4

d. -3

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36 **[Intermediate]** • **Target: Repeated irreducible quadratic factor**

The graph of the quadratic function $f(x) = x^2 - 5x + 6$ is shown below. The graph intersects the x-axis at points A and B. Find the sum of the x-coordinates of points A and B.



a. 5

b. -5

c. 6

d. -6

37 **[Intermediate]** • **Target: Verifying a decomposition by substitution**

The diagram shows the graph of the function $y = f(x)$. The equation of the function is given by $f(x) = ax^2 + bx + c$. Given that the vertex of the parabola is at $(-1, 8)$ and the graph passes through the point $(1, 0)$, find the values of a and b.

a. a = -2, b = -4

b. a = -2, b = 4

c. a = 2, b = 4

d. a = 2, b = -4

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Choosing efficient substitution values • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Consider the quadratic equation $x^2 - kx + 8 = 0$, where k is a real parameter. The graph of the corresponding quadratic function $y = x^2 - kx + 8$ is a parabola. Given that the parabola intersects the x-axis at exactly one point, find the value of k.

a. $-4\sqrt{2}$

b. $-2\sqrt{2}$

c. $2\sqrt{2}$

d. $4\sqrt{2}$

39 [Challenging] • Target: Distinct Linear Factors • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

A random variable X follows a binomial distribution $B(n, p)$. The number of trials is $n = 10$ and the probability of success in a single trial is p. If the mean of X is 4, find the variance of X.

a. 2.4

b. 3.6

c. 6.4

d. 7.2

40 [Challenging] • Target: Repeated Linear Factors • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Find d in the equation $ax + by + cz = d$ of the plane through $A(-1, 3, 4)$, $B(-3, 1, 1)$, and $C(4, 2, -4)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. -59

b. -46

c. -75

d. -58

41 [Challenging] • Target: Irreducible Quadratic Factors • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Find d in the equation $ax + by + cz = d$ of the plane through $A(2, -1, 4)$, $B(2, -3, -3)$, and $C(-4, 4, 2)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. 0

b. -3

c. -6

d. -12

42 [Challenging] • Target: Improper Rational Fractions (Long Division First) • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Find d in the equation $ax + by + cz = d$ of the plane through $A(2, 0, 3)$, $B(-4, 0, 1)$, and $C(0, -2, -3)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. 16

b. 12

c. 24

d. 28

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43 [Challenging] • Target: Distinct linear factors • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Find d in the equation $ax + by + cz = d$ of the plane through $A(3, -1, -2)$, $B(-3, 2, 0)$, and $C(-1, 3, 4)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. 42

b. 32

c. 26

d. 22

44 [Challenging] • Target: Repeated linear factors • Cross-Lesson: 4.02 The Poisson Distribution

Find d in the equation $ax + by + cz = d$ of the plane through $A(-4, -4, 1)$, $B(1, 0, -3)$, and $C(-4, 3, 4)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. -100

b. -60

c. -65

d. -45

45 [Challenging] • Target: Irreducible quadratic factors • Cross-Lesson: 5.01 Further Integration Techniques

Find d in the equation $ax + by + cz = d$ of the plane through $A(3, 1, -3)$, $B(0, -4, 3)$, and $C(3, 3, 0)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. -54

b. -84

c. -63

d. -78

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